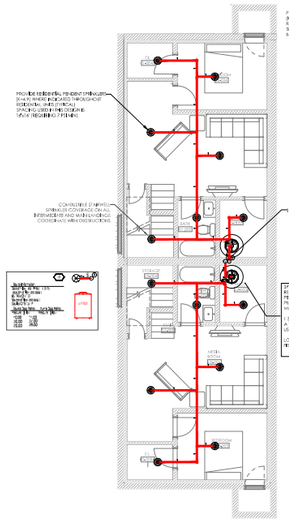
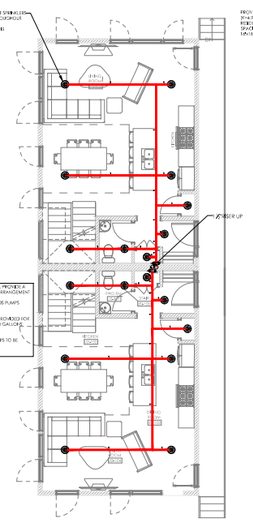
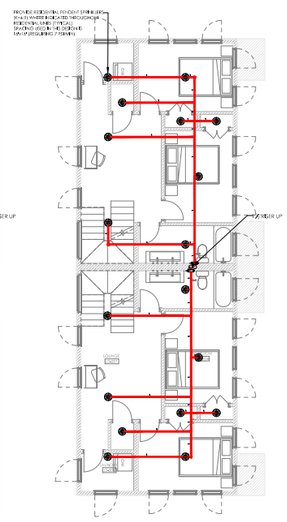


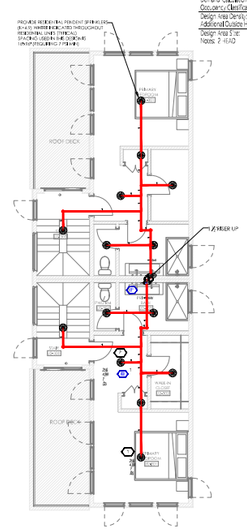
BASEMENT LEVEL SPRINKLER LAYOUT
SCALE: 1/8" = 1'-0"



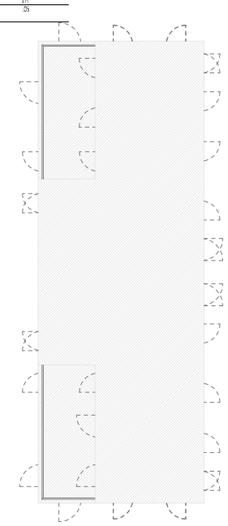
FIRST LEVEL SPRINKLER LAYOUT
SCALE: 1/8" = 1'-0"



SECOND LEVEL SPRINKLER LAYOUT
SCALE: 1/8" = 1'-0"



THIRD LEVEL SPRINKLER LAYOUT
SCALE: 1/8" = 1'-0"



ROOF LEVEL
SCALE: 1/8" = 1'-0"

SPRINKLER PIPE LEGEND

NEW STEEL PIPING
NEW CPVC PIPING
EXISTING PIPING
NEW TAP

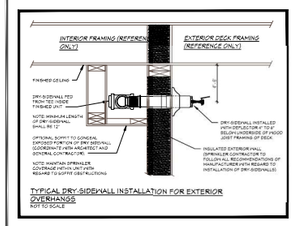
Job: 85 TREMONT ST
Address: 85 TREMONT ST — State: MA — Zip/Postal Code: 02139
City: CAMBRIDGE
Contact #: — Date:
Contractor: CAP DESIGN GROUP — Designer: JK
Address: 335 WASHINGTON ST SUITE 114 — State: MA — Zip/Postal Code: 01801
City: WOBURN
Phone: (617) 644 0014 — Fax: — Email:
Approving Authority: CITY OF CAMBRIDGE
Design Details — Standard: NFPA 13R
Default Sprig Size: 1 — Default Drop Size: 1
Default Sprig Material: CPVC — Default Drop Material: CPVC
Default Sprig Elevation: 0 — Default Drop Elevation: 0

Calculation results for Design Area 1 — TOP FLOOR
This system as shown on: CAP DESIGN GROUP — company print no. — dated 7/3/25
for: 85 TREMONT ST — at: 85 TREMONT ST
contract no. — is designed to discharge at a rate of 0.06 gpm/ft² (L/min/m²) of floor area over a maximum area of 929 ft² when supplied with water at a rate of 26.1 gpm at 26.5 psi at the base of the riser.
Hose stream allowance of 0 is included in the above.
Occupancy classification: LH
Commodity classification:
Maximum storage height:
Storage arrangement:
Number of heads flowing: 2
System Type: Wet
Maximum velocity: 8.8 ft/s

Flow from In-Rack sprinklers:	0 gpm
Flow from Overhead sprinklers:	26.1 gpm
Flow from Inside Hoses:	0 gpm
Flow from Outside Hoses:	0 gpm
Other fixed flows:	0 gpm
Total flow in system piping:	26.1 gpm
Additional flow at/beyond source:	0 gpm
Total of all flows:	26.1 gpm

Pressure Required at Source: 26.5 psi
Pressure Available at Source: 35.7 psi
Surplus Pressure at Source: 9.2 psi

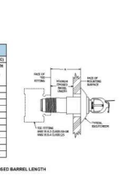
7 DRY SIDEWALL DETAIL



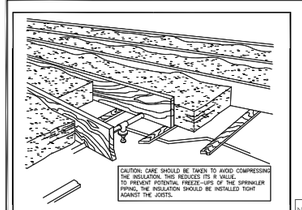
TYPICAL DRY-SIDEWALL INSTALLATION FOR EXTERIOR OVERHANGS
NOT TO SCALE

AMBIENT TEMPERATURE OF PROTECTED AREA OR EXPOSURE SIDE OF THE SPRINKLER	EXPOSED BARREL AMBIENT TEMPERATURE		
	40°F (4°C)	50°F (10°C)	60°F (15°C)
	EXPOSED BARREL LENGTH (MAXIMUM ALLOWABLE DISTANCE FROM THE TOP OF BARREL OR INTERIOR CEILING)		
40°F (4°C)	3	0	0
50°F (10°C)	5	4	0
60°F (15°C)	6 (152)	2	0
70°F (21°C)	8 (203)	7 (178)	0
80°F (27°C)	10 (254)	9 (178)	0
90°F (32°C)	14 (356)	12 (305)	7 (178)
100°F (38°C)	16 (406)	14 (356)	9 (229)
110°F (43°C)	18 (457)	16 (406)	12 (305)
120°F (49°C)	20 (508)	18 (457)	14 (356)
130°F (54°C)	22 (559)	20 (508)	16 (406)

FIGURE 3: EXPOSED BARREL LENGTH



6 FREEZE PROTECTION

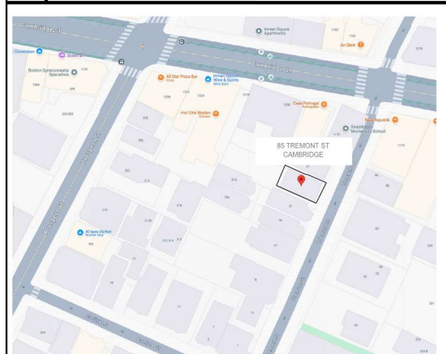


INSULATION PROTECTING SPRINKLER PIPING ON UPPER FLOOR DETAIL
NOT TO SCALE

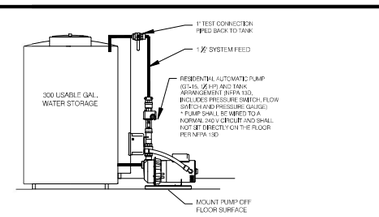
NOTE: ALL COMBUSTIBLE CONCEALED SPACES, INCLUDING INTERSTITIAL SPACES, VOIDS UNDER STAIRS, ETC. WHICH ARE GREATER THAN 6" IN DEPTH, SHALL BE COMPLETELY FILLED WITH NON-COMBUSTIBLE INSULATION BY THE GENERAL CONTRACTOR OR ARE REQUIRED TO BE PROVIDED WITH SPRINKLER PROTECTION IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13 (2013 EDITION). ANY SUCH ARES CREATED DURING CONSTRUCTION SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER IN WRITING BY THE GENERAL CONTRACTOR PRIOR TO INSTALLATION OF GYPSUM BOARD WALLS AND/OR CEILINGS.

NOTE: IT IS THE BUILDING OWNER'S RESPONSIBILITY TO PROVIDE SUFFICIENT HEAT WITHIN ALL AREAS CONTAINING SPRINKLER PIPING TO PREVENT FREEZE-UPS. CAP DESIGN GROUP, INC. TAKES NO RESPONSIBILITY FOR FREEZE-UPS ON THIS SPRINKLER SYSTEM. ANY AREAS OF CONCERN SHALL BE BROUGHT TO THE ENGINEER'S ATTENTION IN WRITING PRIOR TO INSTALLATION.

1 PROJECT INFORMATION



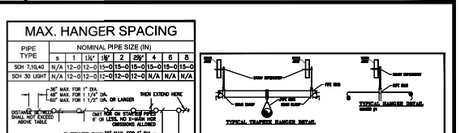
2 PUMP AND TANK DETAIL



3 HANGER DETAIL

MAX. HANGER SPACING

PIPE TYPE	NOMINAL PIPE SIZE (IN)								
	1	1 1/4"	1 1/2"	2	2 1/2"	4	6	8
SCH 7,10,40	N/A	12-0	12-0	15-0	15-0	15-0	15-0	15-0
SCH 30 LIGHT	N/A	12-0	12-0	15-0	12-0	12-0	N/A	N/A



4 GENERAL DETAILS

TABLE 8.6.5.1.2
POSITIONING OF SPRINKLER TO AVOID OBSTRUCTION TO DISCHARGE

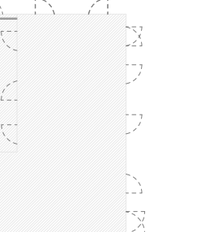
DISTANCE FROM SPRINKLERS TO SIDE OF OBSTRUCTION (A)	MAXIMUM ALLOWABLE DISTANCE OF OBSTRUCTION ABOVE BOTTOM OF OBSTRUCTION (INCHES) (B)
LESS THAN 1 FT	0
1'-0" TO LESS THAN 1'-6"	1'-2"
1'-6" TO LESS THAN 2'-0"	1'-9"
2'-0" TO LESS THAN 2'-6"	2'-6"
2'-6" TO LESS THAN 3'-0"	3'-2"
3'-0" TO LESS THAN 3'-6"	3'-9"
3'-6" TO LESS THAN 4'-0"	4'-2"
4'-0" TO LESS THAN 4'-6"	4'-9"
4'-6" TO LESS THAN 5'-0"	5'-2"
5'-0" TO LESS THAN 5'-6"	5'-9"
5'-6" TO LESS THAN 6'-0"	6'-2"
6'-0" TO LESS THAN 6'-6"	6'-9"
6'-6" TO LESS THAN 7'-0"	7'-2"

NFPA 13R TABLE 8.6.5.1.2 + FIGURE 8.6.5.1.2.
(B) STANDARD PENDENT AND UPRIGHT SPRAY SPRINKLERS

TABLE 8.10.6.1.2
POSITIONING OF SPRINKLER TO AVOID OBSTRUCTION TO DISCHARGE

DISTANCE FROM SPRINKLERS TO SIDE OF OBSTRUCTION (A)	MAXIMUM ALLOWABLE DISTANCE OF OBSTRUCTION ABOVE BOTTOM OF OBSTRUCTION (INCHES) (B)
LESS THAN 1 FT	3
1'-0" TO LESS THAN 1'-6"	1'-10"
1'-6" TO LESS THAN 2'-0"	0'-11"
2'-0" TO LESS THAN 2'-6"	0'-4"
2'-6" TO LESS THAN 3'-0"	0'-1"
3'-0" TO LESS THAN 3'-6"	1'-10"
3'-6" TO LESS THAN 4'-0"	3'-5"
4'-0" TO LESS THAN 4'-6"	5'-0"
4'-6" TO LESS THAN 5'-0"	6'-7"
5'-0" TO LESS THAN 5'-6"	8'-2"
5'-6" TO LESS THAN 6'-0"	9'-11"
6'-0" TO LESS THAN 6'-6"	11'-8"
6'-6" TO LESS THAN 7'-0"	13'-7"
7'-0" AND GREATER	15'-2"

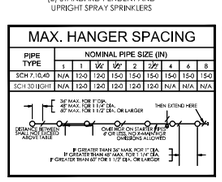
NFPA 13R TABLE 8.10.6.1.2 + FIGURE 8.10.6.1.2
(A) RESIDENTIAL PENDENT AND UPRIGHT SPRAY SPRINKLERS



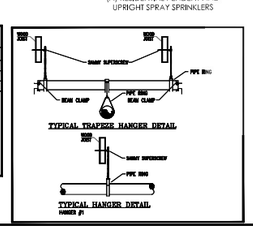
RESIDENTIAL PENDENT DETAIL
NOT TO SCALE

MAX. HANGER SPACING

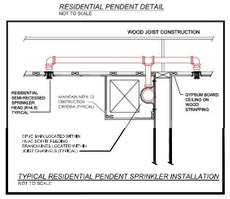
PIPE TYPE	NOMINAL PIPE SIZE (IN)								
	1	1 1/4"	1 1/2"	2	2 1/2"	4	6	8
SCH 7,10,40	N/A	12-0	12-0	15-0	15-0	15-0	15-0	15-0
SCH 30 LIGHT	N/A	12-0	12-0	12-0	12-0	12-0	N/A	N/A



TYPICAL TRAPEZE HANGER DETAIL



TYPICAL HANGER DETAIL



TYPICAL RESIDENTIAL PENDENT SPRINKLER INSTALLATION
NOT TO SCALE

5 GENERAL NOTES

FIRE PROTECTION NOTES:

THE PURPOSE OF THIS FIRE PROTECTION DRAWING AND THE ASSOCIATED FIRE PROTECTION DESIGN NARRATIVE IS TO INDICATE THE PROPOSED MIXED USE RESIDENTIAL SPRINKLER SYSTEM TO BE INSTALLED WITHIN THE RENOVATED MULTI-UNIT BUILDING LOCATED AT **85 TREMONT ST CAMBRIDGE MA.**

THIS BUILDING IS COMPRISED OF A BASEMENT LEVEL WITH THREE RESIDENTIAL FLOOR ABOVE. THE BUILDING IS LESS THAN 30-FT WITH FIRE DEPARTMENT VEHICULAR ACCESS AND THEREFORE DOES NOT REQUIRE A STANDPIPE SYSTEM.

THIS RESIDENTIAL SPRINKLER SYSTEM HAS BEEN DESIGNED IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13D (2013 EDITION) FOR A RESIDENTIAL BUILDING UP TO 3 STORIES IN HEIGHT AND NOT EXCEEDING 12,000 GROSS SF.

THE SPRINKLER CONTRACTOR SHALL FOLLOW THE LATEST REQUIREMENTS OF NFPA 13D (2013 EDITION), MASSACHUSETTS STATE BUILDING CODE (780 CMR) 10TH EDITION (IBC-2021) AND **CAMBRIDGE** FIRE DEPARTMENT REQUIREMENTS.

THIS SYSTEM WILL BE SUPPLIED BY A RESIDENTIAL PUMP AND TANK SYSTEM WHICH HAS BEEN SIZED TO ACCOMMODATE THE FLOW FROM TWO RESIDENTIAL SPRINKLER HEADS FOR A DURATION OF 10-MINUTES IN ACCORDANCE WITH THE REQUIREMENTS OF THE MASSACHUSETTS ADOPTIONS TO THE MASSACHUSETTS STATE BUILDING CODE. THE STORED CAPACITY OF THE WATER TANKS IS 300 TOTAL USABLE GALLONS AND THE PUMP SELECTED IS A GOULDS PUMPS MODEL GT15, 1.5 HP PUMP.

THE SYSTEM HAS BEEN HYDRAULICALLY CALCULATED IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13D, INCLUDING THE FOUR HYDRAULICALLY MOST DEMANDING 2 HEADS IN A SINGLE COMPARTMENT BASED ON THE REQUIREMENTS OF THE SPECIFIC SPRINKLER HEAD AND THE SPACING USED IN THIS DESIGN (16'X16' REQUIRING A MINIMUM OF 13 GPM @ 7.0 PSI). REMOTE AREAS, DENSITIES AND HOSE STREAM ALLOWANCES ARE INDICATED ON THIS DRAWING ALONG WITH SYSTEM DEMANDS AT CONNECTION TO STREET. CALCULATION RESULTS HAVE BEEN COMPARED TO RECENT HYDRANT FLOW TEST INFORMATION OBTAINED FROM THE **CAMBRIDGE** WATER & SEWER DEPARTMENT. CALCULATIONS HAVE BEEN SUBMITTED WITH THIS DRAWING TO THE **CAMBRIDGE** FIRE DEPARTMENT FOR REVIEW.

INSTALLATION NOTES:

ALL WORK SHALL BE PERFORMED BY A MASSACHUSETTS LICENSED SPRINKLER CONTRACTOR. THE SPRINKLER CONTRACTOR SHALL FOLLOW THE LATEST REQUIREMENTS OF NFPA 13D (2013 EDITION), MASSACHUSETTS STATE BUILDING CODE (780 CMR) 10TH EDITION (IBC-2021) AND THE **CAMBRIDGE** FIRE DEPARTMENT.

THE ARCHITECTURAL BACKGROUND OF BUILDING MAY DIFFER SLIGHTLY FROM ACTUAL LAYOUT. DRAWINGS ARE NOT INTENDED TO SHOW ALL OFFSETS AND PIPING ELEVATION CHANGES. CONTRACTOR SHALL FIELD VERIFY ALL MEASUREMENTS PRIOR TO FABRICATION.

CONTRACTOR SHALL HYDROSTATICALLY TEST ALL SPRINKLER PIPING AT 200 PSI FOR 2 HOURS AND IS RESPONSIBLE FOR THE COMPLETION OF ALL ABOVE GROUND TEST CERTIFICATES, SUPPLIED TO THE OWNER.

ALL PIPING INSTALLED THROUGHOUT THE RESIDENTIAL AREAS OF THE BUILDING SHALL BE UL LISTED CPVC SPRINKLER PIPING. ALL PIPING WITHIN THE STORAGE AND UTILITY AREAS OF THE BASEMENT LEVEL WILL BE BLACK STEEL PIPING. ALL PIPING SHALL BE INSTALLED IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13D (2013 EDITION) AND ALL MANUFACTURERS INSTALLATION RECOMMENDATIONS. ALL PIPING SHALL BE PITCHED TO DRAIN WITH LOW-POINT DRAINS AT SECTIONS OF PIPING SUBJECT TO WATER TRAPPING. SPRINKLER CONTRACTOR SHALL PROVIDE PROTECTIVE PLATES WHERE CPVC PIPING IS RUN THROUGH STUDS TO PREVENT PUNCTURING OF THE SPRINKLER PIPING DURING DRYWALL INSTALLATION AS REQUIRED BY NFPA STANDARDS.

ALL SPRINKLER HEADS WITHIN RESIDENTIAL AREAS OF THE BUILDING SHALL BE RESIDENTIAL PENDENT AND SIDEWALL SPRINKLERS. SPRINKLER HEADS WITHIN THE BASEMENT UTILITY AREAS SHALL BE QUICK-RESPONSE COMMERCIAL UPRIGHT SPRINKLERS. ALL HEADS SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S REQUIREMENTS AND THE REQUIREMENTS OF NFPA 13R (2013 EDITION).

BUILDING OWNER IS RESPONSIBLE FOR MAINTAINING THIS SPRINKLER SYSTEM IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 25, INCLUDING THE PROVISION OF HEAT IN ALL AREAS CONTAINING SPRINKLER PIPING AND HEADS TO PREVENT PIPE FROM FREEZING. CAP DESIGN GROUP AND THE ENGINEER OF RECORD TAKE NO RESPONSIBILITY FOR ANY DAMAGES CAUSED BY FREEZE UPS.

COMMONWEALTH OF MASSACHUSETTS
JASON CATALFO
NAMIM
FIRE PROTECTION
NO. 48588
REGISTERED PROFESSIONAL ENGINEER
07/09/2025

FIRE PROTECTION TIER 1

PERMIT NO:	
CONTRACT NO:	
APPROVED BY:	JK
DRAWN BY:	
SCALE:	1/8" = 1'-0"
DATE:	7/9/25
SHEET NO:	FP1 of 1

85 TREMONT ST
CAMBRIDGE, MA

ENGINEER: CAP DESIGN GROUP

CAP DESIGN GROUP
www.capcofire.com
335 Washington St Suite 1114
Woburn, MA 01801
(617)644 0014

SPRINKLER SYMBOLS

SYM	CNT	POSITION	FINISH	TEMP	K	NPT	REF.	NOTES
		U	WHITE	175	4.90	1/2"	V2704	
		RES PEND	WHITE	175	4.90	1/2"	V2740	
		U	WHITE	155	4.4	1/2"	V4431	
		RES SIDE	WHITE	155	4.4	1/2"	V4431	
		RES SIDE	WHITE	155	4.4	1/2"	V4431	

REVISIONS

DESCRIPTION	BY	DATE

NOTES